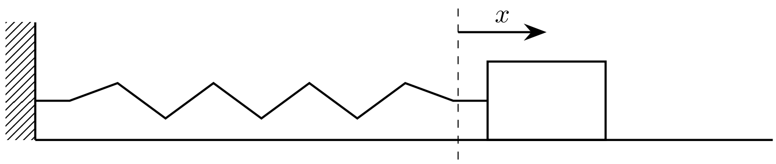
Questions



1. The horizontal displacement of a glider attached to a spring, shown in the diagram above, is modelled by the differential equation

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 16x = 3\sin 4t$$

where x is the displacement, in metres, of the glider from its equilibrium position, t seconds after the motion begins.

(a) (i) Show that a particular solution of the differential equation is

$$x = -\frac{3}{8}t\cos 4t$$

[4]

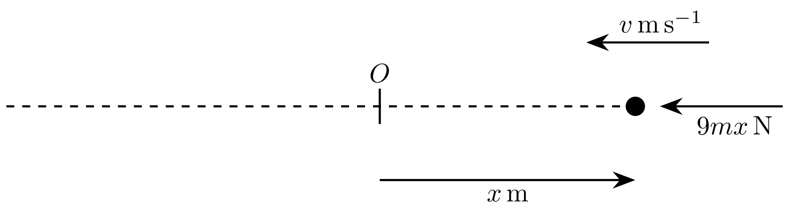
(ii) Hence, find the general solution of the differential equation **[4]**

Initially, the glider is 0.20 m to the right of equilibrium and is moving towards equilibrium at $0.50\,\mathrm{m\,s^{-1}}$. Given that, 4 seconds after the motion begins, the displacement of the glider is β m,

(b) determine, according to the model, the value of β to 3 significant figures **[4]**

(c) Given that the true value of β is 0.46 m, comment on the suitability of the model **[1]**

(d) Write down a differential equation for the glider if the periodic driving force is removed, so that the motion becomes simple harmonic **[1]**



2. A particle of mass m kg moves in a horizontal line through its equilibrium position O. At time t seconds its displacement from O is x metres and its speed is $v\,\mathrm{m\,s^{-1}}$. When the particle is to the right of O, it is acted on by a restoring force of magnitude $9mx$ newtons directed towards O, as shown in the diagram.

At $t = 0$, the particle is 1 m to the right of O and is projected towards O with speed $6\,\mathrm{m\,s^{-1}}$.

(a) In the absence of damping and any external force,

(i) Show that the motion is modelled by the differential equation

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 9x = 0$$

[1]

(ii) State the type of motion **[1]**

(iii) Write down the period of the motion **[1]**

(iv) Find x in terms of t **[4]**

(v) Find the amplitude of the motion **[2]**

(b) In a second model, the particle again starts from the same initial position and velocity, but the motion is damped by a force $6mv$ newtons acting opposite to the direction of motion.

(i) Show that

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 6\frac{\mathrm{d}x}{\mathrm{d}t} + 9x = 0$$

[1]

(ii) State, giving a reason, whether the system is under-damped, critically damped or over-damped **[1]**

(iii) Determine the general solution of this differential equation **[3]**

(c) In a third model, the particle again starts from the same initial position and velocity, and is subject both to the damping force in part (b) and to an external force $25m\cos 4t$ newtons acting in the positive direction. The motion is now modelled by the differential equation

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 6\frac{\mathrm{d}x}{\mathrm{d}t} + 9x = 25\cos 4t$$

(i) Find x in terms of t **[7]**

(ii) In the long term, the particle is observed to perform simple harmonic motion with period about 1.6 seconds. Verify that this is consistent with the answer to part (c)(i) **[2]**

3. The charge q on a capacitor in a driven RLC circuit is modelled by

$$\frac{\mathrm{d}^2 q}{\mathrm{d}t^2} + 2\frac{\mathrm{d}q}{\mathrm{d}t} + 10q = 37\cos 3t$$

where $t \geq 0$ is time.

When $t = 0$, $q = 1$ and $\frac{\mathrm{d}q}{\mathrm{d}t} = 15$.

(a) Find q as a function of t **[9]**

(b) Hence write down an approximate expression for q when t is large, giving your answer in the form $R\cos(3t - \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$ **[4]**

4. The vertical displacement of the body of a car from its steady running height is measured by y metres, where y is positive upwards. The body can move above or below this level, so y can take positive and negative values. After the car leaves a speed hump, the body is momentarily above its steady running height and at rest at time $t = 0$.

A proposed model for the subsequent motion, for $t \geq 0$, is

$$4\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + p\frac{\mathrm{d}y}{\mathrm{d}t} + 9y = 0,$$

where p is a constant and $p \geq 0$.

(a) (i) According to the model, for what value of p would the motion be simple harmonic? **[1]**

(ii) Explain briefly why modelling the motion as simple harmonic is unlikely to be realistic for a car suspension **[1]**

(b) Find the range of values of p for which the model predicts that the body of the car never goes below its steady running height **[2]**

(c) Sketch a possible graph of y against t when p lies outside the range found in part (b) but the motion is not simple harmonic **[1]**

5. A load of mass m kg hangs from a light vertical spring and is free to move only vertically in a liquid.

The spring has spring constant $k\,\mathrm{N\,m^{-1}}$, and the liquid exerts a resistance force of magnitude cv newtons opposite to the direction of motion, where $v\,\mathrm{m\,s^{-1}}$ is the speed of the load.

Let y metres be the downward displacement of the load from its equilibrium position at time t seconds. Assume that $4km > c^2$.

(a) Show that the motion is modelled by

$$m\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + c\frac{\mathrm{d}y}{\mathrm{d}t} + ky = 0$$

and hence show that

$$y = \mathrm{e}^{-\frac{ct}{2m}}\left(P\cos\left(\frac{\sqrt{4km-c^2}}{2m}t\right) + Q\sin\left(\frac{\sqrt{4km-c^2}}{2m}t\right)\right),$$ **[7]**

where P and Q are constants.

(b) It is given that

$$\begin{aligned} m &= 5 \\ c &= 30 \\ k &= 125 \end{aligned}$$

At time $t = 0$, the load is 0.12 m below its equilibrium position and is moving upwards with speed $0.12\,\mathrm{m\,s^{-1}}$.

(i) Find the values of P and Q **[4]**

(ii) Find the first positive value of t for which the load is at its equilibrium position **[5]**

(c) A second model ignores the resistance force but keeps the same values of m and k. Using the same initial conditions as in part (b), find an expression for y in terms of t, giving numerical constants to two significant figures where appropriate **[6]**

6. A manufacturer is testing the response of an electronic weighing platform. The displayed mass, m, measured in kilograms, after a box is placed on the platform is modelled by the differential equation

$$4\frac{\mathrm{d}^2m}{\mathrm{d}t^2} + 4\frac{\mathrm{d}m}{\mathrm{d}t} + 5m = 20$$

where t is the number of seconds after the box is placed on the platform.

(a) Find a general solution of the differential equation **[4]**

Initially, the display reads 0 kg and the displayed mass is increasing at $5\,\mathrm{kg\,s^{-1}}$.

(b) Find, according to the model, the greatest value shown on the display **[8]**

The machine is programmed to print a label only when the displayed mass is less than 4.05 kg.

(c) Determine whether, according to the model, the machine will print a label exactly 8 seconds after the box is placed on the platform **[2]**

7. An analogue meter has a damped pointer. After a short electrical pulse, the angular displacement θ radians of the pointer from its rest position, measured positive clockwise, is modelled by

$$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + 5\frac{\mathrm{d}\theta}{\mathrm{d}t} + 6\theta = 4\mathrm{e}^{-4t}$$

where t is the time, in seconds, after the motion begins.

In one test, at $t = 0$ the pointer is passing through its rest position and has angular velocity $1\,\mathrm{rad\,s^{-1}}$ anticlockwise.

For this test, use the model to

(a) find an equation for θ in terms of t **[8]**

(b) find, to 3 decimal places, the greatest clockwise angular displacement of the pointer from its rest position **[4]**

In this test, the time taken for the pointer to pass through its rest position again was measured as 0.69 seconds.

(c) State, with justification, whether or not this supports the model **[1]**

8. A trolley moves on a horizontal track and is attached to a spring and a dashpot. The trolley is pulled to the right of its equilibrium position and released.

At time t seconds after release, its horizontal displacement is x metres to the right of a fixed point P, where P is the equilibrium position and is 1.2 metres from the left-hand end of the track.

When $t = 0$:

$x = 0.18$,
the velocity of the trolley is $0\,\mathrm{m\,s^{-1}}$,
the acceleration of the trolley is $-0.10\,\mathrm{m\,s^{-2}}$

In the subsequent motion, the displacement of the trolley is modelled by the differential equation

$$3k\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 2k\frac{\mathrm{d}x}{\mathrm{d}t} + 5x = 0, \qquad t \geq 0,$$

where k is a positive constant.

(a) Find the value of k **[2]**

(b) Find the particular solution of the differential equation **[7]**

(c) Hence find, according to the model, the distance of the trolley from the left-hand end of the track 4.5 seconds after it is released **[2]**

(d) State one limitation of the model **[1]**

9. In a feedback system, the error signal E is measured in suitable units.

For $t \geq 0$, E is modelled by the differential equation

$$\frac{\mathrm{d}^2 E}{\mathrm{d}t^2} + \frac{\mathrm{d}E}{\mathrm{d}t} - 6E = -12$$

Initially the error is decreasing at 9 units per second, so $\frac{\mathrm{d}E}{\mathrm{d}t} = -9$ when $t = 0$. It is given that E tends to a finite limit as $t \to \infty$.

(a) Determine the value of E when $t = \ln 2$ **[6]**

(b) Determine the limit of E as $t \to \infty$ **[1]**

10. A machine part executes damped forced vibrations. Its displacement y centimetres from equilibrium at time t seconds is modelled by

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 2\frac{\mathrm{d}y}{\mathrm{d}t} + 10y = 20\cos t + 5\sin t, \qquad t \geq 0$$

(a) Find the general solution of this differential equation. **[6]**

(b) Given that $y = 3$ and $\frac{\mathrm{d}y}{\mathrm{d}t} = 3$ when $t = 0$, find the particular solution describing the motion. **[4]**

(c) The first turning point of this motion for $t > 0$ occurs when $y = a$. Find a correct to 3 significant figures. **[4]**

11. A particle Q moves along the x-axis under a single force. When the time is t seconds, the displacement of Q from the origin is x metres and its velocity is $v\,\mathrm{m\,s^{-1}}$.

The motion of Q is modelled by the differential equation

$$\ddot{x} + \lambda^2 x = 0$$

where $\lambda\,\mathrm{rad\,s^{-1}}$ is a positive constant.

At time $t = 0$, Q is at the point A where $x = -\frac{1}{2}a$, for some positive constant a, and is moving in the positive direction.

Later, Q reaches the point B where $x = a$ and comes to instantaneous rest.

(a) Write down the general solution of this differential equation. **[1]**

(b) Using the answer to part (a), determine expressions, in terms of λ, a and t only, for the following quantities:

(i) x

(ii) v **[3]**

(c) Hence show that

$$v^2 = \lambda^2\left(a^2 - x^2\right)$$ **[1]**

(d) The quantity r is defined by $r = \dfrac{1}{v}$. Using part (c), determine an expression for r_m, the mean value of r with respect to the displacement, as Q moves directly from A to B. **[4]**

(e) One measure of the validity of the model is consideration of the value of r_m. If r_m exceeds 7.2 then the model is considered to be valid. The value of a is measured as 0.240 to 3 significant figures. The value of λ is measured as 0.80 ± 0.02. Determine what can be inferred about the validity of the model from the given information. **[1]**

(f) Find, according to the model, the least possible value of the velocity of Q at time $t = 0$. Give your answer to 2 significant figures. **[2]**

12. An automated inspection trolley of mass 2 kg moves along a straight horizontal rail. The point A is fixed on the rail. The displacement of the trolley from A at time t seconds is x metres, for $t \geq 0$.

For the interval considered, the trolley stays to the right of A and continues to move away from A.

For $t \geq 0$, the trolley is subject to three forces modelled as follows:

- A driving force of magnitude $4(t+1)$ N acts in the positive x-direction.
- A magnetic force of magnitude x N acts towards A.
- A resistive force of magnitude $2\left|\frac{\mathrm{d}x}{\mathrm{d}t}\right|$ N acts opposite to the direction of motion.

(a) Applying Newton's second law gives the differential equation

$$4(t+1) - x - 2\frac{\mathrm{d}x}{\mathrm{d}t} = 2\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$$

(i) Explain the sign of the term involving $\frac{\mathrm{d}x}{\mathrm{d}t}$ in this differential equation **[1]**

(ii) When $t = 0$ the displacement of the trolley is 1 m, and it is moving away from A with speed $1\,\mathrm{m\,s^{-1}}$. Without attempting to solve the differential equation, find the acceleration of the trolley when $t = 0$ **[2]**

(b) Let the particular solution to the differential equation in part (a) be a function g such that $x = g(t)$ for $t \geq 0$. The particular solution can be expressed as a Maclaurin series.

(i) Show that the Maclaurin series for $g(t)$ up to and including the term in t is $1 + t$ **[1]**

(ii) Use your answer to part (a)(ii) to show that the term in t^2 in the Maclaurin series for $g(t)$ is $\frac{1}{4}t^2$ **[1]**

(iii) By differentiating the differential equation in part (a) with respect to t, show that the term in t^3 in the Maclaurin series for $g(t)$ is $\frac{1}{6}t^3$ **[4]**

(c) You are given that the complete Maclaurin series for the function g is valid for all $t \geq 0$. After 0.4 seconds a sensor records that the trolley is 1.45 m from A.

(i) Using the Maclaurin series for $g(t)$ up to and including the term in t^3, evaluate the suitability of the model for determining the displacement of the trolley from A when $t = 0.4$ **[1]**

(ii) Explain why it would not be sensible to use the Maclaurin series for $g(t)$ up to and including the term in t^3 to evaluate the suitability of the model for determining the displacement of the trolley from A when $t = 8$ **[1]**